Do Pakistanis Hold Militant Groups in Low Regard?*

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Abstract

Blair et al. (2013) conclude from endorsement experiments that Pakistanis hold militant groups in "low regard": the groups' endorsements slightly reduce stated policy support, and the authors read the negative sign as sentiment. I reanalyze their data with the measurement model developed for endorsement experiments (Bullock et al., 2011) and find that roughly two in five Pakistanis react positively to a militant group's endorsement (43% averaged across the four groups studied; 95% credible interval 32–53%). The mean reaction is negative but tiny, at most 0.08 ideal-point standard deviations, while the dispersion of reactions is several times larger, so a weakly negative average coexists with widespread favorable reactions. The sign-to-sentiment inference also fails in general: the ATE identifies the balance of switchers, not the proportion of supporters. Explicit behavioral models map the same ATE and baseline to supporter shares from 18% to 99%, and assumption-free bounds place the share of positive reactions between 1% and 58%. Blair et al.'s own subgroup results corroborate the heterogeneity reading: their middle-class treatment effect is indistinguishable from zero. Endorsement experiments can speak to levels of support, but only through explicit behavioral models; the sign of the average treatment effect alone carries almost no information about how many people support a group.

1 Introduction

Do Pakistanis hold militant groups in low regard? Blair et al. (2013) conclude they do, from endorsement experiments in which militant groups' endorsements slightly reduced stated support for policies: the average treatment effects are negative, small, and mostly statistically significant, and the authors read the sign as sentiment. Reanalyzing their data with the measurement model the endorsement-experiment literature itself developed (Bullock et al.,

*https://github.com/finite-sample/endorsement.

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2011), I find that roughly two in five Pakistanis react positively to a militant group's endorsement: 43% on average across the four groups studied (95% credible interval 32–53%), with a mean reaction of at most 0.08 ideal-point standard deviations. A tiny negative mean sits on top of wide disagreement, so "weakly negative on average" and "two in five react positively" are both true. The first fact cannot carry their conclusion: low regard is a claim about the level of support, and the level is what the sign of an average treatment effect (ATE) does not measure.

The inferential gap is general. The ATE from an endorsement experiment is the balance of two forces, supporters of the group raising their stated policy support and opponents lowering theirs. Its sign says which force wins on average, not how many people are on each side. Mapping the ATE to a supporter proportion requires a behavioral model, and the choice matters: explicit models applied to the same two numbers, an ATE of -1.3 percentage points and a baseline of 74%, imply supporter shares anywhere from 18% to 99%. Assumption-free bounds are hardly tighter: any share of positive reactions between 1% and 58% is consistent with the observed response distributions.

Independent evidence points the same way as the model. Blair et al.'s own subgroup estimates put the middle-class treatment effect between -0.6 and 0.1 percentage points, indistinguishable from zero, with the negative average driven by the poor. Bullock et al. (2011), fitting the hierarchical version of the model to these data, find outright positive support in parts of Punjab. And the model-based estimate of 43% lies below the ceiling implied by direct polling once item nonresponse is counted (55% for the Afghan Taliban in Pew's 2012 survey).

Section 2 reviews the design, Section 3 the data, Section 4 the model-based estimates, Sections 5 and 6 what can be inferred without the model, and Section 7 what practitioners should report.

2 Background: Endorsement Experiments

2.1 Design

Endorsement experiments measure latent attitudes toward controversial actors by randomizing whether a group endorses a policy (Bullock et al., 2011). In a standard between-subjects design, respondents see either:

- **Treatment**: "Do you support this policy? [Group Y endorses this policy.]"
- **Control**: "Do you support this policy?"

The identifying assumption is that the endorsement affects policy support only through respondents' attitudes toward the endorsing group. Supporters of the group should increase their policy support; opponents should decrease theirs. The design has been applied to measure support for militant groups in Pakistan (Blair et al., 2013), combatant support in Afghanistan (Lyall et al., 2013), and attitudes in other conflict settings (Blair et al., 2014).

2.2 Standard Interpretation and Its Limits

Researchers typically report the average treatment effect and interpret its sign as indicating "sentiment" toward the endorsing group. A negative ATE is taken to mean respondents view the group unfavorably; a positive ATE suggests favorable attitudes. Blair et al. (2013) exemplify this practice when they conclude that negative coefficients indicate Pakistanis hold militant groups "in low regard."

This interpretation conflates two distinct quantities: the net direction of attitude-driven switching and the proportion of the population holding each attitude. The ATE reflects the net effect of two countervailing forces: supporters increasing their policy support and opponents decreasing theirs. The sign tells us which force dominates on average, but not by how much, and not what proportion of the population falls into each category.

Consider an ATE of -0.01 (a 1 percentage point decrease in policy support). This could arise from 51% opponents each reducing support slightly and 49% supporters each increasing support slightly, or from 1% opponents dramatically reducing support with 99% unaffected, or from many other combinations. Without additional structure, the ATE alone cannot distinguish these scenarios.

2.3 The Identification Problem

The core difficulty is that endorsement experiments are designed to identify the ATE, not the proportion of supporters. Moving from the ATE to a supporter proportion requires a model of individual switching behavior, and different models yield very different answers. This is not a failure of the experimental design; it is a fundamental feature of the inferential problem. The ATE is a well-identified causal quantity. The supporter proportion is a structural parameter that requires additional assumptions.

3 The Experiment and Data

Blair et al. (2013) conducted endorsement experiments in Pakistan measuring attitudes toward four militant groups: the Kashmiri tanzeems, the Afghan Taliban, al-Qa'ida, and the sectarian tanzeems. Respondents were asked about support for four policies (polio vaccination, FCR reform, Durand Line, curriculum standardization) with and without militant group endorsements, on a 5-point scale rescaled to $[0, 1]$. Their survey sampled 6,000 Pakistani adults; the analysis file distributed with the `endorse` R package (Shiraito and Imai, 2022), which I reanalyze, contains the 5,212 respondents with complete responses.[1]

[1] I use the unweighted microdata. Blair et al. weight for oversampling in Balochistan and Khyber Pakhtunkhwa; their weighted Table 1 reports a baseline of 0.796 and an average treatment effect of -0.011, against 0.736 and -0.013 unweighted here. The difference runs in one direction for the analysis below: with the weighted numbers, the binary switching estimate of Section 6 rises from 72% to 79%. Sampling uncertainty is second order throughout: the pooled ATE carries a respondent-clustered standard error of about 0.005, so the width of the ranges reported below reflects model uncertainty, not noise.

Their interpretation has been influential. They write: "We find that Pakistanis in general are weakly negative toward Islamist militant organizations" and "The coefficients are negative and statistically significant at the 10% level for all but the sectarian tanzeems, suggesting that Pakistanis hold militant groups in low regard." The conclusion repeats the claim as the first of their empirical patterns. This follows the standard practice of interpreting the ATE sign as a summary of group-level sentiment. The rest of the paper asks whether that conclusion survives when the mapping from effects to levels is made explicit.

4 Model-Based Estimates of Support

The endorsement-experiment literature has a standard answer to the level question: the measurement model of Bullock et al. (2011), developed on these very data. Each respondent holds a latent position x_i on a policy-support scale, and answers each question according to that position, through question-specific cutpoints (an ordinal probit). An endorsement by group k shifts the respondent's position by a person-specific reaction $s_{ik} \sim \mathcal{N}(\lambda_k, \omega_k^2)$. The mean reaction λ_k is the model's measure of average affect toward group k; the dispersion ω_k measures how much respondents disagree. Because the model pools all four policy questions and uses the full ordinal response distribution, it identifies both parameters, and with them the quantity the "low regard" debate is about: the share of respondents who react positively to group k's endorsement, $P(s_{ik} > 0) = \Phi(\lambda_k/\omega_k)$, which does not depend on the units of the latent scale. I fit the model to the Blair et al. data with the software written by the model's authors (Shiraito and Imai, 2022).[2]

Table 1 reports the results. Averaging over the four groups, 43% of Pakistanis react positively to a militant endorsement (95% credible interval 32–53%). Three groups sit just under one half: 45% for the Kashmiri tanzeems, 47% for al-Qa'ida, and 49% for the sectarian tanzeems. The Afghan Taliban attract the fewest positive reactions, 32%, and the least certainty: their interval runs from 2% to 49% because the estimated dispersion of reactions to them is small and weakly identified. Mean reactions are negative but tiny, between -0.01 and -0.08 ideal-point standard deviations, and only the Afghan Taliban's mean is reliably below zero (posterior probability 0.98). Blair et al.'s own magnitude framing agrees: they describe their strongest effect, for the Afghan Taliban, as "roughly 10% of a standard deviation in support for policies in the control group."

[2]The specification pools questions (a common λ_k across policies), without covariates. Following the protocol of Bullock et al. (2011), I run three parallel chains with overdispersed starting values (200,000 iterations each, 20,000 discarded, every 40th draw kept) and monitor convergence with Gelman–Rubin statistics. All are below 1.02, and the smallest effective sample size across the λ_k exceeds 1,300. Their simulation studies, calibrated to this dataset and its sample size of 5,212, show the model recovers these parameters with little bias and near-nominal coverage.

Table 1. Model-Based Estimates: Share Reacting Positively to Each Group's Endorsement. Mean reaction and reaction SD are in units of ideal-point standard deviations.

Endorser	$P(s > 0)$	95% CI	Mean Reaction	Reaction SD
Kashmiri tanzeems	45%	[25%, 58%]	−0.03	0.25
Afghan Taliban	32%	[2%, 49%]	−0.08	0.20
Al-Qaeda	47%	[40%, 54%]	−0.03	0.49
Sectarian tanzeems	49%	[38%, 57%]	−0.01	0.36
Average	43%	[32%, 53%]	−0.04	0.32

The configuration that produces these numbers is a small negative mean and a much larger dispersion: ω_k runs from roughly 0.2 to 0.5 ideal-point standard deviations, several times $|\lambda_k|$. When reactions are this heterogeneous around a mean this close to zero, "weakly negative on average" and "two in five react positively" are the same fact. The sign of the average is real; the inference from that sign to a population that holds militant groups in low regard is not.

Both source papers corroborate the heterogeneity. Blair et al. (2013) report that the treatment effect for middle-class respondents is between −0.6 and 0.1 percentage points across specifications, indistinguishable from zero; the negative average is driven by the poor, whose effect is −1.9 to −2.6 points (their Table 2). Bullock et al. (2011), fitting the hierarchical version of the model to these data, find clearly positive support in parts of Punjab (three of four groups in Gujranwala) and negative support in Balochistan and NWFP, and no division that suffered more than 50 violent incidents in the prior year shows positive support for any group. An average near zero conceals structured variation; it does not measure consensus.

The alternative specifications imply more positive reactions, not fewer (Table 2). Adding individual covariates (gender, rurality, education, income) moves the average to 44%; replacing the common mean reaction with division-level means, the specification Bullock et al. themselves use, moves it to 49% and tightens every interval. The Afghan Taliban estimate is the most specification-sensitive, rising from 32% to 45% in the hierarchical model.

Table 2. Robustness: Share Reacting Positively Across Model Specifications

Endorser	Baseline	Covariates	Hierarchical
Kashmiri tanzeems	45% [25%, 58%]	45% [34%, 54%]	49% [44%, 55%]
Afghan Taliban	32% [2%, 49%]	39% [27%, 48%]	45% [40%, 51%]
Al-Qaeda	47% [40%, 54%]	46% [40%, 52%]	49% [44%, 54%]
Sectarian tanzeems	49% [38%, 57%]	48% [39%, 56%]	51% [46%, 56%]
Average	43% [32%, 53%]	44% [38%, 51%]	49% [44%, 53%]

5 A Framework for Inference

The estimates above lean on the measurement model's parametric assumptions: reactions are normal, and the ordinal scale identifies their dispersion. This section develops what can be said with less. A reader who accepts only the experimental design learns two numbers, the ATE and the baseline, and this section maps them to supporter proportions under explicit behavioral models. The exercise shows both why the sign-to-sentiment inference fails in general and how wide the honest range is.

5.1 Setup

Consider respondents falling into two categories based on their attitudes toward the endorsing group: supporters (proportion p) who view the group favorably, and opponents (proportion $1-p$) who view it unfavorably. Let δ^+ denote the average treatment effect among supporters and δ^- the average effect among opponents, where we expect $\delta^+ > 0$ and $\delta^- < 0$.

The observed ATE decomposes as:

$$\beta = p \cdot \delta^+ + (1 - p) \cdot \delta^- \tag{1}$$

This equation has three unknowns (p, δ^+, δ^-) and one observed quantity (β). The system is underidentified—a special case of the partial identification problem studied by Manski (1990). Adding the baseline support level (the constant term α from the regression Support $= \alpha + \beta \cdot$ Treatment $+ \varepsilon$) provides a second equation, but the system remains underidentified without restrictions on δ^+ and δ^-.

5.2 Binary Switching

The strongest restriction sets δ^+ and δ^- such that all supporters end up at 1 and all opponents end up at 0 in the treatment group. Under this assumption, the treatment group average is:

$$\text{Treatment average} = p \cdot 1 + (1 - p) \cdot 0 = p \tag{2}$$

Since the treatment average equals $\alpha + \beta$, we get:

$$p = \alpha + \beta \tag{3}$$

The treatment group average equals the proportion of supporters. This result is exact under the binary switching model: the system has three unknowns, one equation from the ATE, and two constraints from binary switching (δ^+ and δ^- are pinned down by the assumption that outcomes are 0 or 1), leaving the system exactly identified.

5.3 Testable Implications of Binary Switching

Binary switching has a strong testable implication. If all treatment group responses are either 0 or 1, the treatment group variance must equal $p(1-p)$. With $p = 0.723$ (the average

treatment mean from the data), the predicted treatment variance is $0.723 \times 0.277 \approx 0.200$, and the standard deviation is approximately 0.45.

Blair et al. (2013) measured policy support on a 5-point scale rescaled to $[0, 1]$. If the treatment group standard deviation is substantially below 0.45, the binary switching model is rejected. The actual treatment standard deviation in the data is approximately 0.29, well below the predicted value and indicating that respondents are not clustering at the extremes of the scale.

This does not invalidate the framework. It means the precise mapping $p = \alpha + \beta$ does not hold, and the analyst must consider softer switching models.

5.4 Softer Switching Models

Without binary switching, the analyst faces the decomposition in Equation (1) with three unknowns. Different assumptions about δ^+ and δ^- yield different values of p.

Symmetric small effects. If $\delta^+ = +d$ and $\delta^- = -d$ for some common effect magnitude d:

$$p = \frac{\beta}{2d} + \frac{1}{2} \tag{4}$$

With $d = 0.05$ (a moderate shift on a $[0, 1]$ scale): $p = -0.013/0.1 + 0.5 = 0.37$, or 37%. The choice of d belongs to the analyst, not the data: $d = 0.02$ gives 18%.

Asymmetric extreme reaction. If opponents react maximally ($\delta^- = -1$, moving to 0) while supporters are unaffected ($\delta^+ = 0$):

$$\beta = -(1 - p) \implies p = 1 + \beta \tag{5}$$

With $\beta = -0.013$: $p = 0.987$, or 99%.

Normal heterogeneity. If individual effects $\delta_i \sim \mathcal{N}(\beta, \sigma^2)$, the proportion with positive effects (loosely, "supporters" in a continuous sense) is:

$$P(\delta_i > 0) = 1 - \Phi\left(\frac{-\beta}{\sigma}\right) \tag{6}$$

With $\beta = -0.013$ and $\sigma = 0.05$: $P(\delta_i > 0) \approx 40\%$. The measurement model of Section 4 is the disciplined member of this family: it estimates the dispersion from the ordinal response distribution instead of assuming it, and arrives at 43%.

These models yield a wide range of implied support—from 18% to 99%—depending on what one assumes about individual switching behavior. Section 6 applies them to the Blair et al. data.

5.5 What Baseline Information Buys

The baseline (constant term) contributes in two ways. First, under binary switching, it provides the second equation needed for exact identification (Equation (3)). Without it,

binary switching says only that $p = \alpha + \beta$ for an unknown $\alpha \in [0, 1]$: any supporter proportion between 0% and $1 + \beta = 98.7\%$ is consistent with the data. Even the strongest behavioral assumption is uninformative about p until the constant pins it down at 72.3%.

Second, even without binary switching, the baseline constrains the plausible range. A control average of 74% means that 74% of respondents support the policy absent the endorsement cue. Any model of switching must be consistent with this fact. Models that imply very low baseline support are ruled out.

The contribution of baseline information is the most practically important finding of this paper. Many analyses of endorsement experiments report only the ATE and its sign. Reporting the treatment group average, or equivalently the constant and the ATE together, enables a much richer set of inferences. The estimate is also robust to partial awareness: if 30% of respondents have no opinion about the endorsing group and are unaffected by the endorsement, the implied support among opinion-holders under binary switching is $\beta/q+\alpha = 72\%$, essentially unchanged.

6 Bounds and Sensitivity

6.1 Binary Switching and the Variance Test

Across all 16 policy × endorser combinations, the average control mean is $\alpha = 0.736$ and the average treatment effect is $\beta = -0.013$, giving an average treatment group mean of 0.723. Baseline support varies across policies: from 67% (Durand Line) to 83% (polio vaccination). Under binary switching, implied support ranges from 71.8% to 72.7% across groups (Table 3). The ordering is stable: the Afghan Taliban draw the most negative ATE and the lowest implied support under every model in the table.

Table 3. Implied Support by Endorser Under Different Behavioral Models

Endorser	ATE	Binary	Symmetric	Normal	Extreme
Kashmiri tanzeems	−0.0113	72.5%	38.7%	41.1%	98.9%
Afghan Taliban	−0.0174	71.8%	32.6%	36.4%	98.3%
Al-Qaeda	−0.0142	72.2%	35.8%	38.8%	98.6%
Sectarian tanzeems	−0.0085	72.7%	41.5%	43.3%	99.2%

But the binary switching model is testable, and the data reject it. Binary switching predicts a treatment group standard deviation of approximately 0.45; the actual treatment standard deviation is 0.29. The variance ratio—observed to predicted—is 0.42 across groups (Table 4). Respondents use the full range of the 5-point scale, not just the endpoints.

Table 4. Variance Test for Binary Switching

Policy	Endorser	Var(T) Obs	Var(T) Pred	Ratio
Polio	Kashmiri tanzeems	0.0573	0.1515	0.38
Polio	Afghan Taliban	0.0638	0.1590	0.40
Polio	Al-Qaeda	0.0596	0.1439	0.41
Polio	Sectarian tanzeems	0.0608	0.1537	0.40
FCR	Kashmiri tanzeems	0.0742	0.2149	0.35
FCR	Afghan Taliban	0.0846	0.2245	0.38
FCR	Al-Qaeda	0.0884	0.2199	0.40
FCR	Sectarian tanzeems	0.0812	0.2142	0.38
Durand	Kashmiri tanzeems	0.1008	0.2228	0.45
Durand	Afghan Taliban	0.0945	0.2194	0.43
Durand	Al-Qaeda	0.1056	0.2255	0.47
Durand	Sectarian tanzeems	0.0985	0.2159	0.46
Curriculum	Kashmiri tanzeems	0.0847	0.1958	0.43
Curriculum	Afghan Taliban	0.0904	0.1938	0.47
Curriculum	Al-Qaeda	0.0853	0.1963	0.43
Curriculum	Sectarian tanzeems	0.0888	0.1990	0.45
Average		0.0824	0.1969	0.42

Researchers can compute this diagnostic from standard regression output without microdata: given β, α, and the treatment-group standard deviation σ, the variance ratio is $\sigma^2/[(\alpha+\beta)(1-\alpha-\beta)]$. A ratio well below 1 rejects binary switching.

6.2 Ceiling Effects on the Discrete Scale

The 5-point scale creates a mechanical asymmetry that biases the ATE downward. With 42% of control responses at 1.0 and 29% at 0.75, supporters who are already at the ceiling cannot shift higher, while opponents at the ceiling can shift down freely. Over two-fifths of the positive-shift potential is lost to truncation.

Formalizing: let each supporter shift up by s categories (capped at 1) and each opponent shift down by s categories (floored at 0). The effective average shifts, accounting for truncation, are $\bar{\delta}^+$ and $\bar{\delta}^-$. Two assumptions do the work here: supporters and opponents share the response distribution observed in the control group, and every respondent shifts the same number of categories. Neither is testable with these data, so the estimates that follow are conditional on both. Solving $\beta = p\bar{\delta}^+ + (1-p)\bar{\delta}^-$ for p, a 1-category shift (± 0.25) yields 59% support despite the negative ATE (Table 5). Larger assumed shifts imply more support (66% at two categories, 70% at three) because truncation bites harder on the up side. The ceiling effect explains part of why the ATE is negative: supporters' positive reactions are mechanically attenuated by the scale.

Table 5. Ceiling Effects on the 5-Point Scale

Shift	Eff. δ^+	Eff. δ^-	Ceiling Loss	Implied p
1 category	0.145	-0.236	0.105	58.6%
2 category	0.218	-0.454	0.282	65.6%
3 category	0.251	-0.631	0.499	70.1%

6.3 Implied Support Across Models

Table 6 collects implied support under all models considered, from symmetric effects with $d = 0.02$ (18%) to extreme opponent reaction (99%). Figure 1 and Figure 2 display the range graphically. The two symmetric rows make the menu dependence concrete: shrinking the arbitrary effect size d from 0.05 to 0.02 cuts the implied supporter share from 37% to 18%, and nothing in the data disciplines the choice of d.

Table 6. Implied Support Under Different Behavioral Models

Model	Key Assumption	Implied Support
Extreme opponent reaction	$\delta^- = -1$, $\delta^+ = 0$	98.7%
Binary switching	All supporters $\to 1$, opponents $\to 0$	72.3%
5-point scale, 2-step shift	Shift ± 2 categories, truncated	65.6%
5-point scale, 1-step shift	Shift ± 1 category, truncated	58.6%
Normal heterogeneity ($\sigma = 0.05$)	$\delta_i \sim \mathcal{N}(\beta, 0.05^2)$	39.9%
Symmetric effects ($d = 0.05$)	$\delta^+ = +0.05$, $\delta^- = -0.05$	37.2%
Symmetric effects ($d = 0.02$)	$\delta^+ = +0.02$, $\delta^- = -0.02$	17.9%

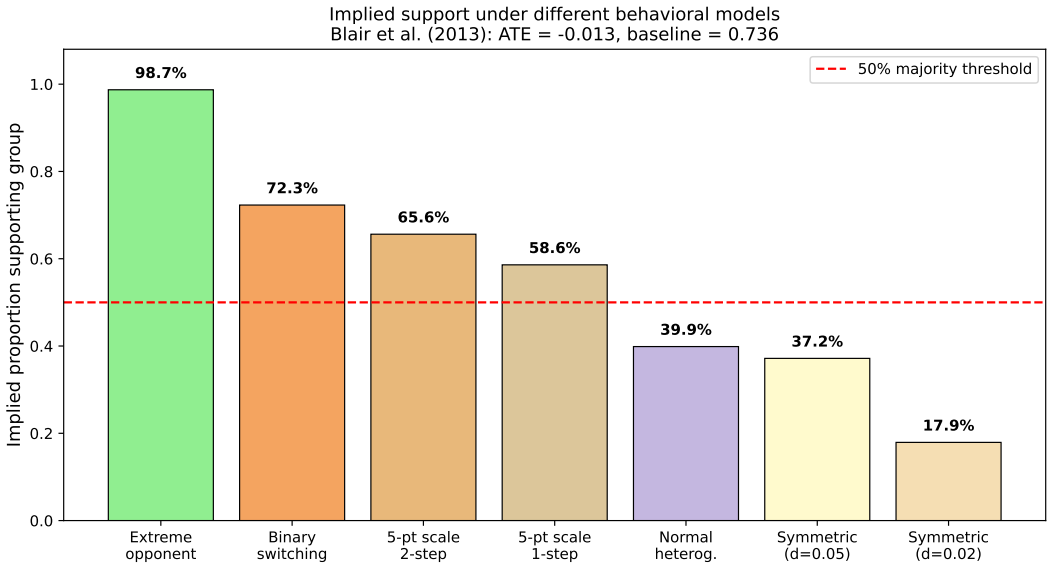


Figure 1. Implied support varies substantially across behavioral models. No single estimate is privileged without a defended choice of model.

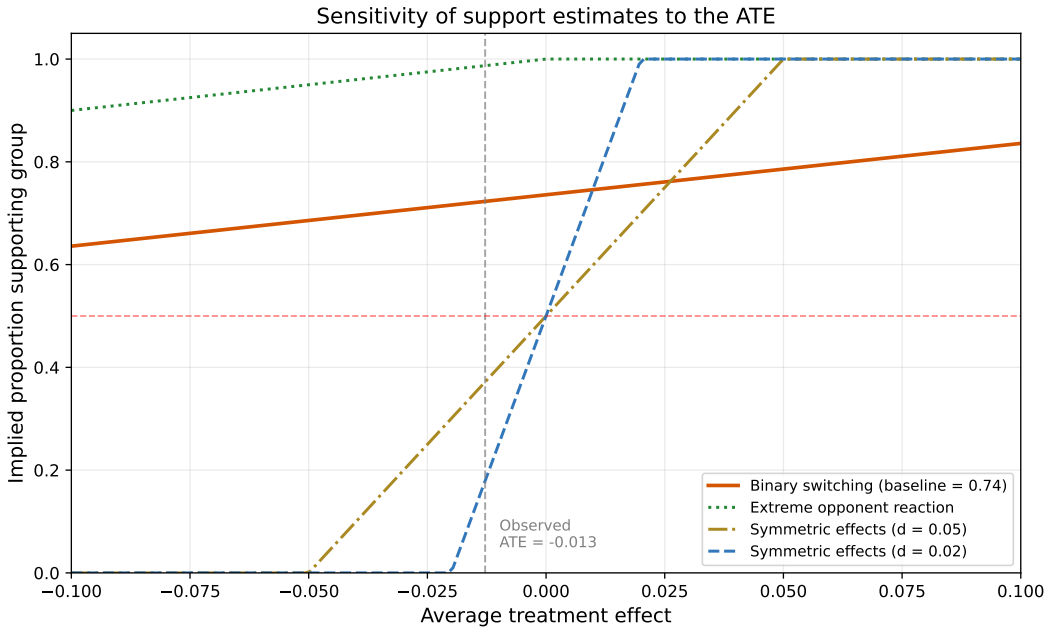


Figure 2. Sensitivity of support estimates to the ATE under different models.

The width of this range—18% to 99%—reflects the genuine underidentification of the problem. The data constrain the supporter proportion only loosely without a commitment to a specific behavioral model, and the range includes low support: a claim that endorsement experiments refute "low regard" would itself require a defended model.

A model-free calculation shows exactly how little the data alone pin down. The experiment identifies the two marginal distributions of responses, treated and untreated, but

not the joint distribution linking them, and any pairing (coupling) of the two marginals is consistent with the data. Across all such pairings, the share of respondents whose response the endorsement strictly raises can be as low as 1% or as high as 58% (Table 7). Randomly pairing treatment and control responses gives intermediate figures—the treated response is higher 34% of the time, lower 37%, and tied 29%—but only the interval $[1\%, 58\%]$ is a bound. The interval spans the majority threshold: the data are consistent with almost no one reacting positively to the endorsement and with a substantial majority doing so. The bounds are nearly identical across the four groups.

Table 7. Random-Pairing Probabilities and Coupling Bounds on the Share of Positive Reactions

Endorser	$P(T < C)$	$P(T = C)$	$P(T > C)$	Min. Positive	Max. Positive
Kashmiri tanzeems	37%	29%	34%	2%	58%
Afghan Taliban	38%	29%	34%	1%	58%
Al-Qaeda	37%	29%	34%	1%	58%
Sectarian tanzeems	36%	29%	35%	1%	58%
All groups	37%	29%	34%	1%	58%

6.4 External Validation

Direct polling on militant group support in Pakistan yields much lower figures. Pew Research Center surveys from 2012–2014 find 8–14% favorable views of the Taliban and al Qaeda, with substantial fractions declining to offer an opinion—21% to 41% depending on the group and year (Pew Research Center, 2012, 2013, 2014). The gap between these figures and the endorsement experiment estimates (18–72% depending on model) is large.

Part of this gap plausibly reflects social desirability bias and fear, which is precisely the motivation for indirect measurement. Blair et al. report item nonresponse of 6% to 22% on direct questions about militant groups in their survey, against 0.6% to 7.6% on the endorsement questions, and they note don't-know rates around 40% on Pew's direct al-Qa'ida and Taliban favorability questions in 2008. But even assigning every nonrespondent to the support camp yields at most roughly 55% (14% favorable plus 41% no opinion for the Afghan Taliban in 2012). This ceiling is below the binary switching estimate of 72%, close to the discrete-scale estimate of 59%, and comfortably above the model-based estimate of 43%. The external evidence does not decisively adjudicate between models, but it is most consistent with the middle of the range, where the measurement model of Section 4 sits.

7 Discussion

Four recommendations follow from this analysis.

Report level implications, not only ATEs. The measurement model of Section 4 turns the same data into a share of positive reactions with a credible interval. Reporting

$P(s > 0)$ alongside the ATE answers the question readers actually ask, and the software to do so is mature (Shiraito and Imai, 2022).

Report treatment group averages, not just ATEs. Under binary switching, the treatment average directly estimates the supporter proportion. Even when binary switching is implausible, the treatment average provides a useful anchor. Reporting only the ATE and its sign discards information.

Do not equate the ATE sign with "sentiment." The sign tells you which switching force dominates, not whether the group is popular or unpopular. With high baseline support, negative ATEs can coexist with majority support under multiple behavioral models.

Discuss identifying assumptions explicitly. Moving from ATEs to support levels requires a behavioral model. Authors should state which model they assume, defend it, and report how sensitive results are to alternatives. The variance test provides a simple diagnostic for the binary switching assumption.

8 Conclusion

Blair et al. (2013) conclude from small negative treatment effects that Pakistanis hold militant groups in "low regard." Their own data, run through the measurement model the endorsement-experiment literature built for this purpose, say something more uncomfortable: roughly two in five Pakistanis react positively to militant group endorsements (43%, credible interval 32–53%), because a mean reaction of at most 0.08 ideal-point standard deviations coexists with dispersion several times larger. "Weakly negative on average" is accurate. "Low regard," as a description of the population, is not supported.

The reduced-form analysis explains why no reading of the sign alone could have settled this. The ATE identifies the balance of switchers, not the proportion of supporters: explicit behavioral models map the same ATE and baseline to supporter shares from 18% to 99%, and assumption-free bounds place the share of positive reactions between 1% and 58%. The measurement model's answer is one point in that set, distinguished by assumptions that are explicit and conventional, normal reactions with a dispersion identified from scale usage, rather than true by construction. Two caveats apply: the interpretation of the constant as baseline support requires a correctly specified regression, and every estimate assumes the endorsement manipulation is credible—if respondents doubt that militant groups endorse benign policies, the measured reactions understate true attitudes.

Endorsement experiments identify average treatment effects well but are underidentified for the quantity researchers most want: the proportion of supporters. Moving from effects to levels requires explicit behavioral assumptions. Researchers should report the level implications of their estimates under such models, report treatment group averages and baselines, and treat the sign of the ATE as what it is: the direction of the average, not the distribution of the population.

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